

BSP implementation guide

Electricity Market Service - FFR capacity market

Business process: Balancing capacity market

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Important Information

This implementation guide is relevant only for the new market platform for FFR which is expected to be launched in June (testing) and October (production) 2026

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1 Introduction

1.1 Background

Electricity Market Service (EMS) has been developed to facilitate amongst others the FFR capacity market, with purpose to ensure the availability of reserves in to ensure operational security.

This document describes FFR reserve capacity market.

Fast Frequency Reserves (FFR) are types of Balancing Services that TSOs use for balancing of the Nordic synchronous area. In DK2, FFR balancing capacity is procured in one auction and satisfies a proportion of a common reserve requirement. The auction takes place after the day-ahead market.

1.2 Scope

This document covers the technical implementation details on the BSP side, when integrating with Electricity Market Service (EMS) by describing the processes required to interact with the FFR capacity market. Both functional and technical aspects are covered. The intended audience of this document are the participating BSPs. If the BSP role has not been separated from the BRP role, the target audience for this document are the BRPs providing balancing services. Thus, the acronym BSP can be looked as synonymous to BRP in these cases.

The main processes described are:

- Reserve requirement publication process
- Bid submission process
- Market result publishing process

1.3 Terms and definitions

Acronym	Term	Definition
ACE	Area Control Error	
BSP	Balancing Services Provider	<i>A market participant with reserve-providing units or reserve-providing groups able to provide balancing services to TSOs</i>
BRP	Balance Responsible Party	<i>A market participant or its chosen representative responsible for its imbalances</i>
CIM	IEC Common Information Model	
ECP	Energy Communication Platform	<i>Reference implementation of MADES standard</i>
EMS	Electricity Market Service	<i>Electricity Market Service is the platform that Energinet uses to facilitate all FFR auction communication between TSO and BRP/BSP.</i>
FFR	Fast Frequency Reserves	<i>The FFR that can be activated by an automatic control device designed to reduce the Frequency Restoration Control Error (FRCE) / ACE</i>
MADES	Market Data Exchange Standard	<i>Communication IEC standard designed by ENTSO-E</i>
MOL	Merit Order List	<i>A list of Balancing Energy bids sorted in order of their bid prices, used for the activation of Balancing Energy bids within a Coordinated Balancing Area.</i>
MTU	Market Time Unit	<i>The period for which the market price is established or the shortest possible common time period for the two bidding zones, if their market time units are different.</i>
TSO	Transmission System Operator	<i>A party that is responsible for a stable power system operation (including the organisation of physical balance) through a transmission grid in a geographical area. In the Nordic synchronous area, there are four TSOs: Svenska kraftnät, Fingrid, Energinet.dk and Statnett.</i>

1.4 References

- Ref [1] [Energy Market Communications – Acknowledgement Business Process and Contextual Model for CIM European Market \(IEC 62325-451-1\)](#)
- Ref [2] [ENTSO-E Reserve Bid document – UML model and schema v 7.6](#)
- Ref [3] [ENTSO-E Balancing Document – UML model and schema v 4.5](#)
- Ref [4] [ENTSO-E Reserve Allocation Result document – UML model and schema v 6.4](#)
- Ref [5] [ENTSO-E Acknowledgement document – UML model and schema v. 8.1](#)

2 Business context

2.1 FFR market timelines

The FFR capacity market involves several operational phases. The diagram below shows the timelines for the FFR capacity market.

- involved: Bidding zone DK2

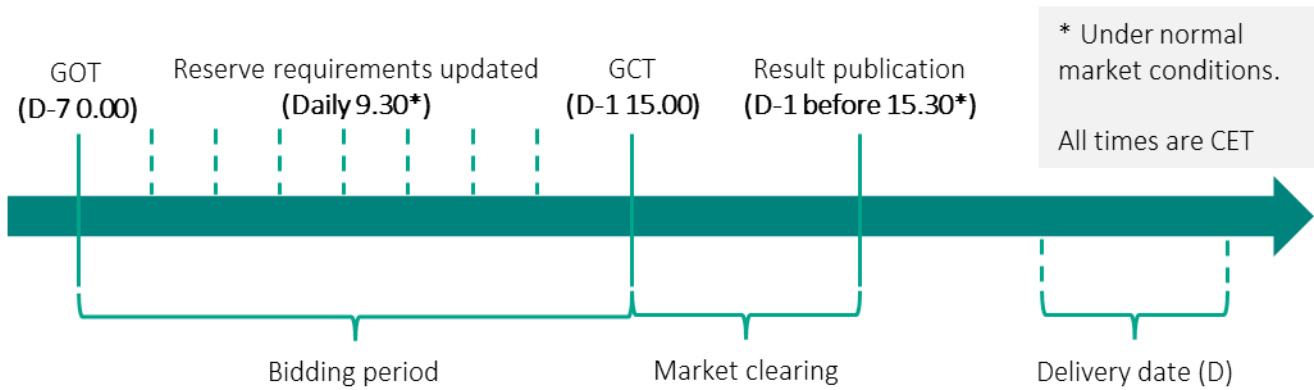


Figure: FFR market timeline

2.2 Timeline description

- FFR market gate opening time: BSPs can submit their from D-7 00:00 CET.
- FFR reserve requirements are published daily from D-7 up to D-1 on energidataservice.dk and are visible in EMS.
- Subsequently, when reserve requirements are updated, EMS republishes them through the described reserve requirement publishing process to BSPs up until the gate closure time (GCT).
- Between the capacity market Gate Opening Time (GOT) and the capacity market Gate Closure Time (GCT), the BSPs can submit and update their bids (Bidding period).
- BSPs can submit their bids to the FFR market until D-1, 15:00 CET.
- After gate closure, the market clearing runs. The Bid Selection Optimisation is executed, and the market results are published to BSPs. FFR result publication is no later than D-1, 15:30 CET
- If full requirement is not satisfied, Energinet has the possibility to procure additional FFR. Updated market results with updated marginal price will then be published later.

2.3 System context

This document provides detailed information about the message exchanges between EMS and the BSPs. The other exchanges are shown for contextual information purposes only and are outside the scope of this document.

The table below provides an overview of the flows between EMS and the BSPs. The table shows that IEC CIM is used as data exchange format and ECP/ EDX is used as communication platform. Please refer to chapter 5 for more details related to ECP/EDX.

Sender	Receiver	Data	Document	ECP message type
EMS	BSP	Publish reserve requirements	iec62325-451-7-reservebiddocument.xsd – version 7.6	MO-MCC-RESERVE-REQUIREMENTS-V7-6
BSP	EMS	Bids	iec62325-451-7-reservebiddocument.xsd – version 7.6	MO-MCC-BIDS

EMS	BSP	Acknowledgement	iec62325-451-1-acknowledgementdocument.xsd – version 8.1	"original message code"-ACK
EMS	BSP	Accepted contracts	iec62325-451-7-reserveallocationresult.xsd – version 6.4	MO-MCC-ACCEPTED-BIDS-BSP-V6-4
EMS	TSO	Market result	iec62325-451-6-balancing.xsd – version 4.5	MO-MCC-MARKET-RESULTS-V4-5

Table 1 Flow between EMS and BSP

2.4 General rules

The FFR capacity market business process flows assume certain basic rules. These rules are described below.

Date and time

Date and time are expressed using the standard XML format for date and time: YYYY-MM-DDTHH:MM:ssZ, formatted using the universal time standard **UTC** by adding a 'Z' behind the time - like this: 2022-06-14T22:00:00Z

Document coverage

The beginning and ending date and time of the period covered by the document shall cover just one CET/CEST day.

Daylight saving time

- In **winter**, the period is from 23:00 UTC to 23:00 UTC
- In **summer**, the period is from 22:00 UTC to 22:00 UTC
- On the date of the change from winter to summer time, the period is from 23:00 UTC to 22:00 UTC. This change occurs on the last Sunday in March at 01:00 UTC
- On the date of the change from summer to winter time, the period is from 22:00 UTC to 23:00 UTC. This change occurs on the last Sunday in October at 01:00 UTC

Document identification and revision number

The document identification must be unique over time for the sender in question. Furthermore, the document identification itself should not have any significant meaning. The revision number is not used and shall always be equal to '1'.

Update/cancel principles

In general, a new received document will always completely replace a previously received document. **Update** of any time series is done by sending a new document honouring these rules

- A new document mRID (document identification)
- The same revision number (always equal to '1')
- A newer created date-time
- The same *period/day, domain, auction.mRID and subject market participant* (for bids) as for the data being updated

I.e. to **cancel** one or more time series, a new document is submitted honouring the above rules, omitting the bids that should be cancelled.

Cancel all bids

In order to cancel all bids by a given sender, and day, a new document should be submitted with a dummy Bid_TimeSeries using element "status" and value A09 = cancelled. The time series must include the element auction.mRID, with the valid market code.

- The document period is specified as usual using the reserveBid_Period.timeInterval element.
- The document's scope (bidding zone DK2) is specified using the domain.mRID element.
- The dummy timeseries must include the status element and a reference to the market using element auction.mRID.
- The content of the other mandatory timeseries elements is ignored.

Working example:

```
<ReserveBid_MarketDocument>
```

...

```
<Bid_TimeSeries>
```

```
  <mRID>8c520815945a43e88e1fb72e9ad05507</mRID>
```

```
  <auction.mRID>FFR_CAPACITY_MARKET_DK</auction.mRID>
```

```
  <businessType>B74</businessType>
```

```
  <acquiring_Domain.mRID codingScheme="A01">10Y1001A1001A91G</acquiring_Domain.mRID>
```

```
  <connecting_Domain.mRID codingScheme="A01">10YDK-2-----M</connecting_Domain.mRID>
```

```
  <quantity_Measure_Unit.name>MAW</quantity_Measure_Unit.name>
```

```
  <currency_Unit.name>EUR</currency_Unit.name>
```

```
  <price_Measure_Unit.name>MAW</price_Measure_Unit.name>
```

```
  <divisible>A02</divisible>
```

```
  <status>
```

```
    <value>A09</value>
```

```
  </status>
```

```
  <flowDirection.direction>A01</flowDirection.direction>
```

```
  <marketAgreement.type>A01</marketAgreement.type>
```

```
  <Period>
```

```
    <timeInterval>
```

```
      <start>2000-01-01T00:00Z</start>
```

```
      <end>2000-01-01T01:00Z</end>
```

```
    </timeInterval>
```

```
    <resolution>PT60M</resolution>
```

```
    <Point>
```

```
      <position>1</position>
```

```
      <quantity.quantity>0</quantity.quantity>
```

```
      <price.amount>0</price.amount>
```

```
    </Point>
```

```
  </Period>
```

```
</Bid_TimeSeries>
```

```
</ReserveBid_MarketDocument>
```

Acknowledgement

For the *Bid submission* process, an acknowledgement document, as defined in Ref [5] is generated as response, either completely accepting the received document or rejecting it completely. Partial acceptance of documents is not used. The codes and values used in this document are provided in chapter 4.10

The acknowledgement will always contain a document level Reason, with either code A01 (Message fully accepted) or A02 (Message fully rejected).

The acknowledgement may also contain further document level Reason(s), explaining document level errors.

The acknowledgement may also contain one or more Rejected_Timeseries, with one or more timeseries level Reason(s).

The acknowledgement may also contain one or more InError_Period within the Rejected_Timeseries, with one or more Reason(s) for a given time interval.

See example code snippet below:

```
<Acknowledgement_MarketDocument>
```

```
...
```

```
<Rejected_TimeSeries>
  <mRID>4CDF6AAA-4C0D-98DB-94CDE58FB4B5</mRID>
  <InError_Period>
    <timeInterval>
      <start>2021-01-07T00:00Z</start>
      <end>2021-01-07T01:00Z</end>
    </timeInterval>
    <Reason>
      <code>A59</code>
      <text>Quantities across bids exceeds BSPs portfolio.</text>
    </Reason>
  </InError_Period>
  <Reason>
    <code>A22</code>
    <text>Invalid BSP</text>
  </Reason>
</Rejected_TimeSeries>
<Reason>
  <code>A02</code>
  <text>Document fully rejected.</text>
</Reason>
<Reason>
  <code>A51</code>
  <text>The attribute createdDateTime cannot be in the future.</text>
</Reason>
<Acknowledgement_MarketDocument>
```

Time series period

Within a time series period, the position must always begin with '1' and be incremented by '1' for each subsequent position, forming a series of contiguous numbers covering the complete range of the period. Furthermore, *gaps* in the time series are allowed. In this case, the time series will comprise of several periods to indicate the gaps.

See code snippet below:

```
<Bid_TimeSeries>
```

```
...
```

```

  <Period>
    <timeInterval>
      <start>2018-09-17T03:00Z</start>
      <end>2018-09-17T05:00Z</end>
    </timeInterval>
    <resolution>PT60M</resolution>
    <Point>
      <position>1</position>
      <quantity.quantity>3.2</quantity.quantity>
      <price.amount>25.20</price.amount>
    </Point>
    <Point>
      <position>2</position>
      <quantity.quantity>2.1</quantity.quantity>
      <price.amount>25.20</price.amount>
    </Point>
  </Period>
  <Period>
    <timeInterval>
      <start>2018-09-17T15:00Z</start>
      <end>2018-09-17T17:00Z</end>
    </timeInterval>
    <resolution>PT60M</resolution>
    <Point>
      <position>1</position>
      <quantity.quantity>1.6</quantity.quantity>
      <price.amount>25.20</price.amount>
    </Point>
    <Point>
      <position>2</position>
      <quantity.quantity>1.6</quantity.quantity>
      <price.amount>25.20</price.amount>
    </Point>
  </Period>
</Bid_TimeSeries>

```

3 Business process

This chapter provides information related to the business rules that apply.

3.1 Sequence diagram

The sequence diagram for the whole FFR market is depicted below. Note that the first and last shown exchanges are provided for information only and are outside the scope of this document.

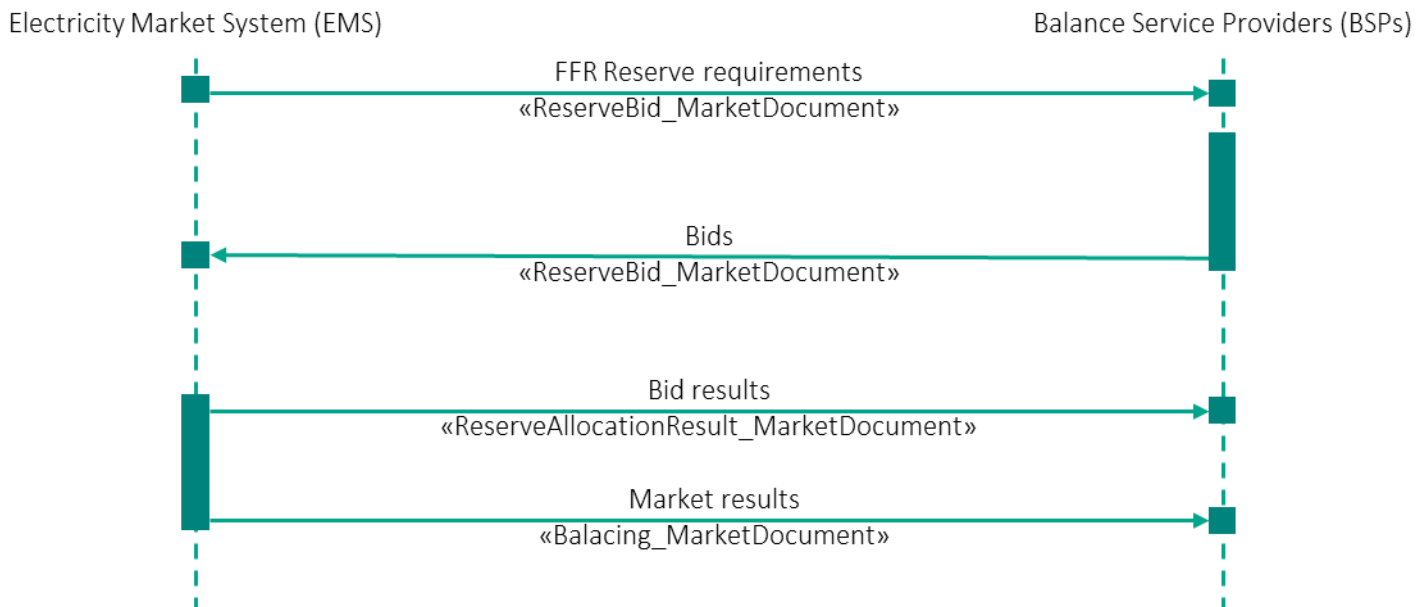


Figure: Sequence diagram for the FFR capacity market

3.2 Reserve requirements

EMS publishes reserve requirements as CIM xml messages when the reserve requirements are submitted to the EMS. Reserve requirements are updated daily until Gate Closure Time, and EMS will always publish the updated reserve requirements to all BSPs with known ECP configurations.

The reserve requirements are defined and shared only for DK2.

Reserve requirements for the FFR include only 1 time series, covering the following item:

- **Minimum regulation (volume to be procured in bidding zone DK2)**

The minimum regulation that is defined as a part of the reserve requirements definition on bidding zone DK2, work as a demand. In this case, the bid selection will try to select bids to satisfy the minimum regulation volume from the bids in bidding zone DK2.

EMS sends out reserve requirement changes if the requirements change. The published reserve requirements are daily data in hourly resolution. They apply to a single auction and delivery day only.

3.3 Bid submission

During the bidding phase, the participating BSPs provide all information related to bids for FFR capacity market. The *ReserveBid_MarketDocument* is used to provide this information. Alternatively, bids can be submitted via the EMS web user interface. The same business rules apply for both alternatives.

To be able to submit bids to EMS, the BSP needs to have a valid **Portfolio** in the market in DK2. A portfolio represents that the BSP has prequalified resources to provide the FFR balancing services in DK2 and an agreement with Energinet to participate in the market.

For a general description of the *ReserveBid_MarketDocument* schema, please refer to Ref [2]. The codes and values to be used in this document are provided in chapter 4.

The following business rules apply to the bid submission process.

3.3.1 General rules

- A Reserve Bid Document contains a set of bids (a bid is represented by a time series).
- A Reserve Bid Document shall be for only one *BSP*.
- The ECP endpoint used to send the data must be associated with the subject party either by the subject party being the same as the sender and being directly associated with endpoint used. (EMS is not supporting the subject party having an active agent-principal relationship.)
- The delivery period for a bid must belong to the same tender period for all bids in a Reserve Bid Document
- Bids are given only for bidding zone DK2, that means as portfolio-bids. The number of bids submitted may not exceed the maximum number of bids set per portfolio.

3.3.2 Bid quantity

- Bid quantity may vary for all hours in the time series
- Bid quantity time series does not have to be a continuous interval of values. Gaps, where the quantities are not specified, are allowed and there is no need to specify zero quantities instead of gaps. The only condition is that at least one quantity must be specified in the bid.
- Bid quantity must be in the interval between minimum and maximum bid sizes as specified in "Energinet's Udbudsbetingelser" available on Energinet's website.
- Bid quantity must be a multiple of 0,1 MW.
- Bid quantity may not exceed the maximum bid quantity according to the registered BSP portfolio.

3.3.3 Bid price

- Bid price must be present and have the same value for all positions in a time series, i. e. there is just one price for the whole bid. It is therefore not necessary to split the quantity time series into multiple bids if the price is the same for all time intervals. However, If BSP needs to set different prices for different time intervals, multiple bids must be created.
- Bid price must be in the interval minimum 0,00 EUR to maximum 9999,99 EUR
- Bid price must be a multiple of 0,01 EUR.

3.3.4 Bid divisibility

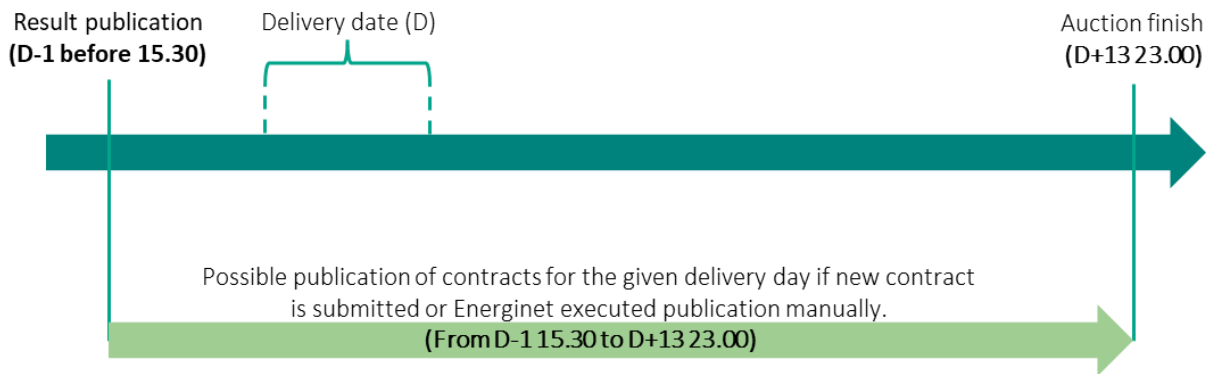
In the FFR market, all bids are considered indivisible on an MTU level. Consequently, the accepted quantity is equal to the offered quantity or zero on the MTU level.

3.4 Contracts submission

The contract generally represents the capacity contract between BSP and Energinet. Energinet is able to submit Contracts on behalf of the BSPs to the EMS. The submission of the contracts is possible once the market auction has been cleared. The BSP is informed about their own contracts via corresponding data flow (see chapter 4.8). For more information about Contracts please visit the chapter 0

3.5 Market result publishing

FFR market results publishing timeline



FFR market results overview

After the bid selection optimization process is completed, and the result from the auction is ready, the following information is published from the EMS system:

Document		Information	Timing
Accepted bids (auction contracts) <i>ReserveAllocationResult_MarketDocument</i>		Only own bids	• When bid selection optimization process is completed and the auction results have been published.
		Both procured and offered prices and quantities	
Market result <i>Balancing_MarketDocument</i>	Volumes	Market volumes	• Each time a new contract is registered, or contract publication is manually executed by MOP (accepted price included).
	Prices	Market prices	
Contracts (contracts of all types) <i>ReserveAllocationResult_MarketDocument</i>		Only own contracts	

Settlement of accepted uses the pay-as-cleared methodology, meaning that the market result includes clearing prices for DK2 and market time unit with the price being included both in the balancing market document and the reserve allocation result market document.

Contracts

The *ReserveAllocationResult_MarketDocument* as defined in 4.8 is used to provide information about BSPs contracts. Firstly, a status of the bids (Auction contracts) that submitted to the auction is provided after auction clearing. One of the following reasons is used to indicate the status of the Auction contract (bids):

- bid is accepted (**A73**)
- bid is not accepted (**B09**)
- bid has been divided to permit acceptance (**A72**)
- Tender unavailable in MOL list (the bid is rejected by TSO) (**B16**)

EMS can also contain another type of Contracts which may be placed to the system by Energinet after publication of the auction results.

The other types of contracts are:

- Auction (default for the standard bids created for the FFR auction)

- Procurement outside market (direct trade outside the auctions)
- Return, buy-back of the BSP's obligation to Energinet by the BSP. Return can be placed into the EMS after the auction clearing.
- Announced non-delivery (non-delivery of the contract announced ex-post by BSP)
- Unannounced non-delivery (non-delivery of the contract not announced by BSP but found out by Energinet)

Main features of the contracts:

- Auction contracts (standard bids created for the auction)
 - Has offered quantity
 - Can be positive or negative-
 - Negative offered quantity can only be a result of contracts created outside the auction (production reduction, planned maintenance, unplanned outages).
 - Has accepted quantity (zero in case of non-accepted bid) when offered quantity exists
- Can be positive or negative in special case of the negative bids.
 - Has offered price
 - Has accepted price when non-zero accepted quantity exists
 - Can be different from hour to hour
- Can be the same or higher than offered price
 - Is equal to the clearing price in case of the positive accepted quantity
- Non-auction contracts:
 - Has no offered quantity
 - May have accepted quantity
 - Positive in case of the *Procurement outside market*, ~~Transfer swap contracts~~
 - Negative in case of the *Return*, *Announced non-delivery*, *Unannounced non-delivery*.
 - Have no offered price
 - Must have accepted price when non-zero accepted quantity exists
 - Accepted price can be differed from hour to hour
 - *Procurement outside market* has price confirmed outside of the market
 - *Return* has price equal to Maximum product price at the moment of the Return submission
- *Announced non-delivery* has price equal to Maximum product price * Penalty for announced non-delivery fixed each hour at the specified time (*visible at Auction timeline, attribute Close procurements*)
- *Unannounced non-delivery* has price equal to Maximum product price * Penalty for unannounced non-delivery fixed each hour at the specified time (*visible at Auction timeline, attribute Close procurements*)

Note: Maximum product price is calculated as maximum clearing price including prices of the FFR bids accepted of the type of *Procurement outside market*.

The BSPs receive information about their own contracts. EMS distributes one *ReserveAllocationResult_MarketDocument* per BSP and market result publication run.

Order of result publication runs:

- All **auction** contracts after auction clearing
- All contracts (**volumes and price**) after contract registration by Energinet

Market results – volumes and prices

The *Balancing_MarketDocument* as defined in 4.9 is used to provide information about the market prices and the total volume per hour and direction. The codes and values to be used in this document are provided in chapter 4.9. EMS distributes one *Balancing_MarketDocument* message per BSP and market result publication run. Market result messages are shared with all BSPs which are registered with known ECP integration.

Republishing of market results

EMS can republish the market results multiple times due to different reasons:

- If the auction result is valid but cannot be processed by recipients due to inconsistencies in party identifiers. In this case, master data in EMS requires to be updated before the market result is republished.
- Republishing can also be initiated when communication with some or all recipient endpoints has previously failed.
- If the auction result is valid and bids covered less volume than the requirement, Energinet might decide to procure additional volumes after the normal clearing time.

To support the update/cancel principle described in o and ensuring that the new market result replaces any earlier received market results, all documents that are sent will have new document mRID and updated createdDateTime.

Finish with non-purchase

Finish with non-purchase is a special market result, where

- All reason codes in the accepted bids message are set to **B09** – Bid not accepted
- All accepted volumes in the total market results are set to 0

There are three possible scenarios for such zero results to be distributed:

- There are no reserve requirements
- No auction result has been published within the final publication deadline
- The auction has been cancelled because of extraordinary market conditions or a permanent failure during market clearing

Non-purchase market results completely replace and invalidate earlier published results.

4 Data flow descriptions

This chapter provides the dependencies for the documents used to support the FFR capacity market.

4.1 Classification

The dependency matrices describe the classification of the attributes. The following classifications are used:

- M – The information is mandatory, i.e. the element is mandatory in the XSD schema
- R – The information is required, i.e. the element is not mandatory in the XSD schema but is required by the business process
- D – The information is dependent, i.e. the presence depends on certain condition(s)
- O – The information is optional.

4.2 Coding schemes

When communication with EMS, the coding scheme to identify senders and receivers can vary. Energinet defines which coding scheme and party code to use for the market participants. EMS use fixed codes and coding scheme A01.

Code	Coding Scheme	Description and reference
I	EIC	Energy Identification Codes: https://www.entsoe.eu/data/energy-identification-codes-eic/
A10	GS1/ GLN	Global Location Number, provided by GS1: https://gepir.gs1.org/index.php/search-by-gln
NDK	Danish national	

4.3 Bidding zones

The FFR capacity market consists of the following bidding zones:

Name	TSO	mRID	Coding scheme
DK2	Energinet	10YDK-2-----M	EIC

4.4 Control areas

The control area for Denmark (DK) is: **10Y1001A1001A796**, however it is not used because the scope of this document is only bidding zone DK2.

4.5 Macro areas

The scope of this document is only the FFR market in DK2, and no other bidding zones or macro areas.

4.6 Publishing of reserve requirements from EMS

The data flow is used for distribution of the reserve of bidding zone DK2 towards BSPs.

The publishing of reserve requirements using the ReserveBid_MarketDocument in version 7.6., please refer to Ref [2]

Document scope:

Each received document shall always contain data for one day.

ReserveBid_MarketDocument		451-7-reservebiddocument.xsd – version 7.6
mRID	M	Unique identification of the document.
revisionNumber	M	Constant value of 1
Type	M	B21 = Reserve need document
process.processType	R	Z14 = Fast frequency reserve (FFR)
sender_MarketParticipant.mRID	M	10X1001A1001A248 (EMS)
		A01 = EIC coding scheme
sender_MarketParticipant.marketRole.type	M	A34 = Reserve Allocator
receiver_MarketParticipant.mRID	M	Identification of the party receiving the document
		Identification is supported by several coding schemes. See Chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	A46 = Balancing Service Provider (BSP)
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
reserveBid_Period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ
domain.mRID	M	EIC identification of the market area: 10Y1001A1001A91G = Nordic Area
		A01 - EIC coding scheme

BidTimeSeries**Send DK2 requirements**

The following time series are created for DK2:

- demand
 - value represents in-elastic demand,
- ~~minimum regulation,~~
- ~~maximum regulation~~

mRID	M	Unique identification of the need timeseries.
auction.mRID	R	Constant value: FFR_CAPACITY_MARKET_DK
businessType	M	For reserve requirements: Relevant for DK2 in the acquiring domain. B75 (Need) = The requirement that should ideally be covered in the specified amount.
acquiring_Domain.mRID	M	10YDK-2-----M
		A01 - EIC coding scheme
		EIC identification of the market area:

connecting_Domain.mRID	M	10Y1001A1001A91G (Nordic Market Area) A01 - EIC coding scheme
quantity_Measurement_Unit.name	M	MAW = megawatt
Divisible	M	A01 = Yes
flowDirection.direction	M	A01 = Up

Period		
timeInterval	M	Period covered (in ISO 8601 UTC format). Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ
Resolution	M	PT60M – the time resolution. Must equal the duration of the timeInterval.

Point		
Position	M	Sequential value beginning with 1.
quantity.quantity	M	Quantity of in-elastic demand for DK2, for business type B75

4.7 Bid updates to EMS

The data flow is used by BSPs to submit the bids to the capacity market. The ReserveBid_MarketDocument in version 7.6 is used for Bids updates, please refer to Ref [2].

Document scope

Each received document shall always contain FFR data for one day and one BSP within bidding zone DK2. The scope is defined by combination of the elements:

- reserveBid_Period.timeInterval,
- subject_MarketParticipant.mRID (= subject BSP),
- domain.mRID (= bidding zone DK2),
- auction.mRID of first time series in the document (= FFR market).

The bids within the scope which are already stored in the EMS are deleted and replaced by new bids in the incoming XML file. In other words, each new XML document always completely replaces all existing bids in the scope (day, subject BSP, bidding zone DK2, FFR market).

Consequently, the submission of the empty XML document would lead to the removal of all existing bids in the scope. However, the empty document does not allow to identify the market in question because the element auction.mRID is on time series level. In that case, the sender must send one "dummy" time series with element status and value A09 = cancelled. The only purpose of this time series is to identify the market. The content of the time series is ignored. The completely empty document will be rejected with the reason code A69 and reason text: „At least one time series must be present“.

ReserveBid_MarketDocument

451-7-reservebiddocument.xsd – version 7.6

mRID	M	Unique identification of the document
revisionNumber	M	Constant value of "1"

Type	M	B40 = Complete Reserve Bid Document
process.processType	R	Z14 = Fast frequency reserve (FFR) Identification of the party sending the document = valid BSP
sender_MarketParticipant.mRID	M	----- Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
sender_MarketParticipant.marketRole.type	M	A46 = Balancing Service Provider (BSP) 10X1001A1001A248 (EMS)
receiver_MarketParticipant.mRID	M	----- A01 = EIC coding scheme
receiver_MarketParticipant.marketRole.type	M	A34 = Reserve Allocator Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ Used for the general validation that a newer document (i. e. with a more recent createdDateTime) in the same scope has not already been processed.
reserveBid_Period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ Start and end interval must define an entire CET Day 10YDK-2-----M
domain.mRID	M	----- A01 - EIC coding scheme Identification of the party responsible for the bid subject_MarketParticipant must be:
subject_MarketParticipant.mRID	R	• the same as the BSP as the sender_MarketParticipant. ----- Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
subject_MarketParticipant.marketRole.type	R	A46 = Balancing Service Provider (BSP)

Bid_TimeSeries

mRID	M	Unique identification of the time series
auction.mRID	R	Constant value: FFR_CAPACITY_MARKET_DK

businessType	M	B74 = The time series provides an offer to provide reserves.
acquiring_Domain.mRID	M	10Y1001A1001A91G (Nordic Market Area) A01 - EIC coding scheme
connecting_Domain.mRID	M	10YDK-2-----M A01 - EIC coding scheme
quantity_Measure_Unit.name	M	MAW = megawatt
currency_Unit.name	R	EUR = euro
price_Measure_Unit.name	R	MAW = megawatt
divisible	M	A02 = Indivisible <i>No reduction possible on the quantity</i> The information about the status of the bid. The element is optional. If the element is present, it must contain the value A09 (Cancelled) and is used for being able to cancel all bids for a given delivery date and market. In that case, the content of the time series is ignored.
status	O	If a time series with such a status is part of the document, it must be the only time series in the document. The time series is present only for the purpose of the identification of the market (which is part of the document scope). As a result of uses of the status element, all bids provided by given BSP (subject_MarketParticipant.mRID) will be deleted from the system.
flowDirection.direction	M	A01 = Up

Period

		Period covered (in ISO 8601 UTC format)
timeInterval	M	Time interval must be a subset of reserveBid_Period.timeInterval and must not overlap with any other time interval of the same time series. Start: YYYY-MM-DDTHH:MMZ End: YYYY-MM-DDTHH:MMZ Constant value PT1H or PT60M
resolution	M	The precision of the interval that the different points within the time series cover is <i>one hour</i>

Point

position	M	Position within the time interval. Sequential value beginning with 1
quantity.quantity	M	Offered quantity
		The price of the product offered
price.amount	M	Must be specified as the same value across all points across all periods of the bid.

4.8 Publishing contracts results from EMS

The data flow is used for distribution of detailed results (accepted and not accepted bids = Auction contracts) towards BSPs after auction clearing. The data flow is also used for distribution of detailed results of all other contracts towards BSPs (see also chapter o).

Only the BSP who submitted at least one bid in the DK2 FFR market receives the document. Message is not sent to BSPs without relevant bids.

Description of the document:

The publishing of detailed results (accepted and not accepted bids) using the ReserveAllocationResult_MarketDocument in version 6.4, please refer to Ref [4].

ReserveAllocationResult_MarketDocument		iec62325-451-7-reserveallocationresult.xsd – version 6.4
mRID	M	Unique identification of the document
revisionNumber	M	Constant value of " 1 "
type	M	A38 = Reserve allocation result document
process.processType	R	Z14 = Fast frequency reserve (FFR)
sender_MarketParticipant.mRID	M	10X1001A1001A248 (EMS) ----- A01 = EIC coding scheme
sender_MarketParticipant.marketRole.type	M	A34 = Reserve Allocator ----- Identification of the party receiving the document
receiver_MarketParticipant.mRID	M	----- Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	A46 = Balancing Service Provider (BSP) ----- Date and time of document creation (in ISO 8601 UTC format)
createdDateTime	M	YYYY-MM-DDTHH:MM:SSZ
reserveBid_Period.timeInterval	M	Period covered (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ
domain.mRID	M	10Y1001A1001A91G (Nordic Market Area) ----- A01 - EIC coding scheme

TimeSeries

Note for time series:

Each time series represent one contract. If the logical receiver of the data flow is BSP, all existing contracts of this BSP are included.

mRID	M	An identification that uniquely identified the time series
bid_Original_MarketDocument.mRID	R	Information extracted from the bid document
bid_Original_MarketDocument.revisionNumber	R	Constant value of "1"
bid_Original_MarketDocument.bid_BidTimeSeries.mRID	R	The identification of the time series that was used in the original tender - the reference to the bid
bid_Original_MarketDocument.tendering_MarketParticipant.mRID	R	The ID of the tendering party Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
auction.mRID	R	Possible values: FFR_CAPACITY_MARKET_DK B95 = Procured capacity (used for Auction contracts) A02 = Internal trade (used for Procurement outside market contract) C11 = Production reduction (used for Capacity return contract) A53 = Planned maintenance (used for Announced non-delivery contract) A54 = Unplanned outage (used for Unannounced non-delivery)
businessType	M	10Y1001A1001A91G (Nordic Market Area) A01 - EIC coding scheme
acquiring_Domain.mRID	M	10YDK-2-----M A01 = EIC coding scheme
connecting_Domain.mRID	M	A01 = EIC coding scheme
marketAgreement.type	R	A01 = daily
marketAgreement.mRID	R	Constant value of "NA"
quantity_Measurement_Unit.name	M	MAW = megawatt
currency_Unit.name	R	EUR = Euro
price_Measurement_Unit.name	R	MAW = megawatt
flowDirection.direction	M	A01 = Up

Reason (within Time Series)

code	M	Information whether the bid has been accepted in standard way or due to the bottleneck (forcibly accepted bid). B49 = contract has not been accepted due to bottleneck
------	---	---

Note: acceptance due to bottleneck is not currently possible in the FFR market

Information whether the bid (Auction contract) has been accepted or not.

- **If businessType = B95 and code = B49**

A73 = Bid in question has been accepted (full acceptance of the bid in all hours of the day)

A72 = The original bid quantity has been divided to enable it to be accepted (partial acceptance of the bid, at least in one hour).

B09 = Bid not accepted

B16 = Tender unavailable in MOL list (the bid is rejected by TSO)

- **If businessType ≠ B95 not populated**

Note: relevant only for bids (Auction contract)

A reason for rejection described by in the words.
Only present if reason code = B16

Period

Period covered (in ISO 8601 UTC format)

YYYY-MM-DDTHH:MMZ

Each time interval represents one continuous interval of hours where accepted quantity is **available**.

1	2	3	4	5	6	7	8	9	Etc.
V	V	V	V			V	V		Etc.

"V" represents any filled-in accepted quantity.

Note:

Bids (Auction contracts, businessType = B95)

- In a standard auction, the accepted quantity is **available** for each hour for which there is a bid quantity. The accepted quantity is always filled with at least zero (in case of non-accepted quantity).
- For forcibly acceptance is accepted quantity **available** only for hours where forcibly acceptance is greater than zero.

Non auction contracts (businessType ≠ B95)

- For non-auction contracts, the accepted quantity is available only for hours where acceptance is greater than zero.

In this case, the following intervals are built:

1–4 (i. e. 00:00–4:00)

7–8 (i. e. 6:00–8:00)

PT60M – the precision of the interval that the different points within the time series cover is *one hour*

Point

position	M	Position within the time interval. Sequential value beginning with 1
quantity	M	The accepted quantity of the contract identified for a point <i>Description of the relationship between the quantity and other elements:</i> Reason.code = B49 and businessType = B95 quantity of the bid (auction contract) accepted in standard auction and: • ReasonCode = A73 => (quantity must be higher than 0) and (quantity must be equal to secondaryQuantity) for all points of the TS (the bid is accepted as a whole) • ReasonCode = A72 => quantity must be ≥ 0 and at the same time quantity must be $<$ secondaryQuantity in at least one point of the TS • ReasonCode = B09 or B16: quantity is = 0 (for all points of the TS) The accepted price of the contract identified for a point <i>Note: Price.amount is present if quantity > 0</i>
price.amount	D	The accepted price can be different from the original offered price and can be different hour by hour in the case that market operates within the Pay as clear method or non-auction contract is presented. The offered quantity identified for a point
secondaryQuantity	R	For all supported ReasonCodes: secondaryQuantity is present secondaryQuantity is not present of non-auction contracts (businessType $\neq$ B95) The offered price identified for a point
bid_Price.amount	R	For all supported ReasonCodes: bid_Price.amount is present bid_Price.amount is not present of non-auction contracts (businessType $\neq$ B95)

4.9 Publishing total market results from EMS

The data flow is used to distribute overall auction results (area volumes and prices) towards BSPs. The document is sent to all BSPs with known ECP configurations. The document contains the data of all bidding zones in the given market.

The **new** publishing of reserve requirements using the Balancing_MarketDocument in version 4.5., please refer to Ref [3].

Balancing_MarketDocument		iec62325-451-6-balancing.xsd – version 4.5
mRID	M	Unique identification of the document
revisionNumber	M	Constant value of "1"
type	M	B34 = Market result document
process.processType	M	Z14 = Fast frequency reserve (FFR)
sender_MarketParticipant.mRID	M	10X1001A1001A248 (EMS) A01 = EIC coding scheme
sender_MarketParticipant.marketRole.type	M	A34 = Reserve Allocator Identification of the BSP receiving the document
receiver_MarketParticipant.mRID	M	Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	M	A46 = Balancing Service Provider (BSP)
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
area_Domain.mRID	R	10Y1001A1001A91G (Nordic Market Area) A01 – EIC coding scheme
period.timeInterval	M	The period covered by the document (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ

TimeSeries		
mRID	M	An identification that uniquely identifies the time series
businessType	M	C17 = Market price and volume
acquiring_Domain.mRID	R	10Y1001A1001A91G (Nordic Market Area) A01 - EIC coding scheme
connecting_Domain.mRID	R	10YDK-2-----M A01 - EIC coding scheme
type_MarketAgreement.type	R	A01 = daily

flowDirection.direction	R	A01 = Up
currency_Unit.name	R	EUR = Euro
quantity_Measurement_Unit.name	R	MAW = megawatt
price_Measurement_Unit.name	R	MAW = megawatt
auction.mRID	R	Possible values: FFR_CAPACITY_MARKET_DK

Period

timeInterval	M	Period covered (in ISO 8601 UTC format) YYYY-MM-DDTHH:MMZ
Resolution	M	PT60M = the precision of the interval that the different points within the time series cover is one hour

Point

Position	M	Position within the time interval. Sequential value beginning with 1
Quantity	R	Total volume procured
procurement_Price.amount	R	Market price

4.10 Acknowledgement document returned by EMS

Acknowledgement_MarketDocument	iec62325-451-1-acknowledgement.xsd – v 8.1	
mRID	M	Unique identification of the document
createdDateTime	M	Date and time of document creation (in ISO 8601 UTC format) YYYY-MM-DDTHH:MM:SSZ
sender_MarketParticipant.mRID	M	10X1001A1001A248 A01 = EIC coding scheme
sender_MarketParticipant.marketRole.type	M	A34 = Reserve Allocator Identification of the BSP receiving the ACK
receiver_MarketParticipant.mRID	M	Identification is supported by several coding schemes. See chapter 4.2: "Coding schemes"
receiver_MarketParticipant.marketRole.type	R	A46 = Balancing Service Provider (BSP)
received_MarketDocument.mRID	R	Information extracted from the bid document
received_MarketDocument.revisionNumber	R	Information extracted from the bid document
received_MarketDocument.createdDateTime	R	Information extracted from the bid document

Rejected_TimeSeries

mRID	D	Identification of the rejected time series
Reason	D	One or more of the reasons described below A01 = Message fully accepted A02 = Message fully rejected A05 = Sender without valid contract. Used if the BSP is not eligible for the given product in the current bid area A57 = Deadline limit exceeded/Gate not open A59 = Not compliant to local market rules Other reason codes may also be used.
code	M	A specific reason for the rejection can be specified and included here. E.g. «Time series ID»: Quantities use incorrect number of decimals
text	M	

5 ECP

ECP is the integration channel that is supported for communication with the EMS market platform. It is used for all data exchanges between EMS and BSPs. Information that describes how to install and get started with ECP is found here: <https://energinet.dk/el/elmarkedet/aktor-pa-balancemarkedet/kom-i-gang-med-ecp/>.

5.1 ECP/EDX for capacity market

Which message types to use

To see list of the ECP message types used for EMS please visit chapter 2.3.

Which service to use

The addressing convention used for communicating in the ECP network is *SERVICE-service.code*.

Note: EMS does not validate ECP service name but it is recommended to use values described in the following table.

Market	Market code	ECP Service name
FFR capacity market Denmark	FFRCAP_DK	C

5.2 How it works

ECP delivers messages from the sender to a recipient within single ECP network. Messages transported through the ECP network can be any text or binary data. IEC CIM is used as data exchange format. Alongside with the message, ECP transfers also message metadata. These are (among others) information about sender and recipient. The former is used by platform to authenticate the message sender. ECP consists of three main components: *endpoint*, *component directory* and *broker*. See figure below:

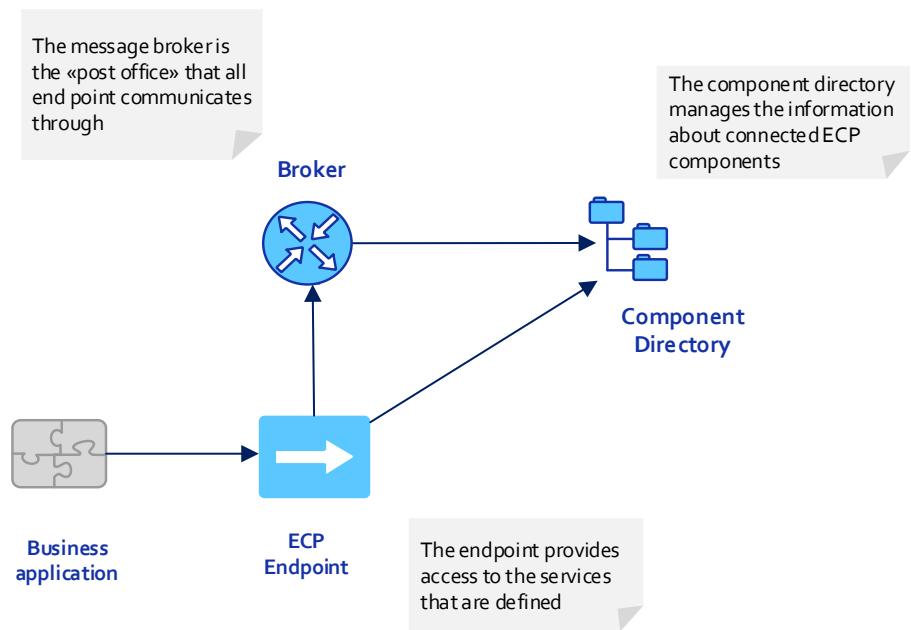


Figure 1 ECP Main Components

5.3 EDX

EDX is an extension to ECP, and is used to define the network configuration, and introduces the concept of services, service providers and consumers. The two central parts of the EDX network is the service catalogue and the toolbox. An EDX network consists of multiple toolboxes and a single service catalogue. These components communicates via ECP and is responsible for distribution of the network configuration. Messaging occurs directly between the toolboxes. See figure below for EDX network overview:

